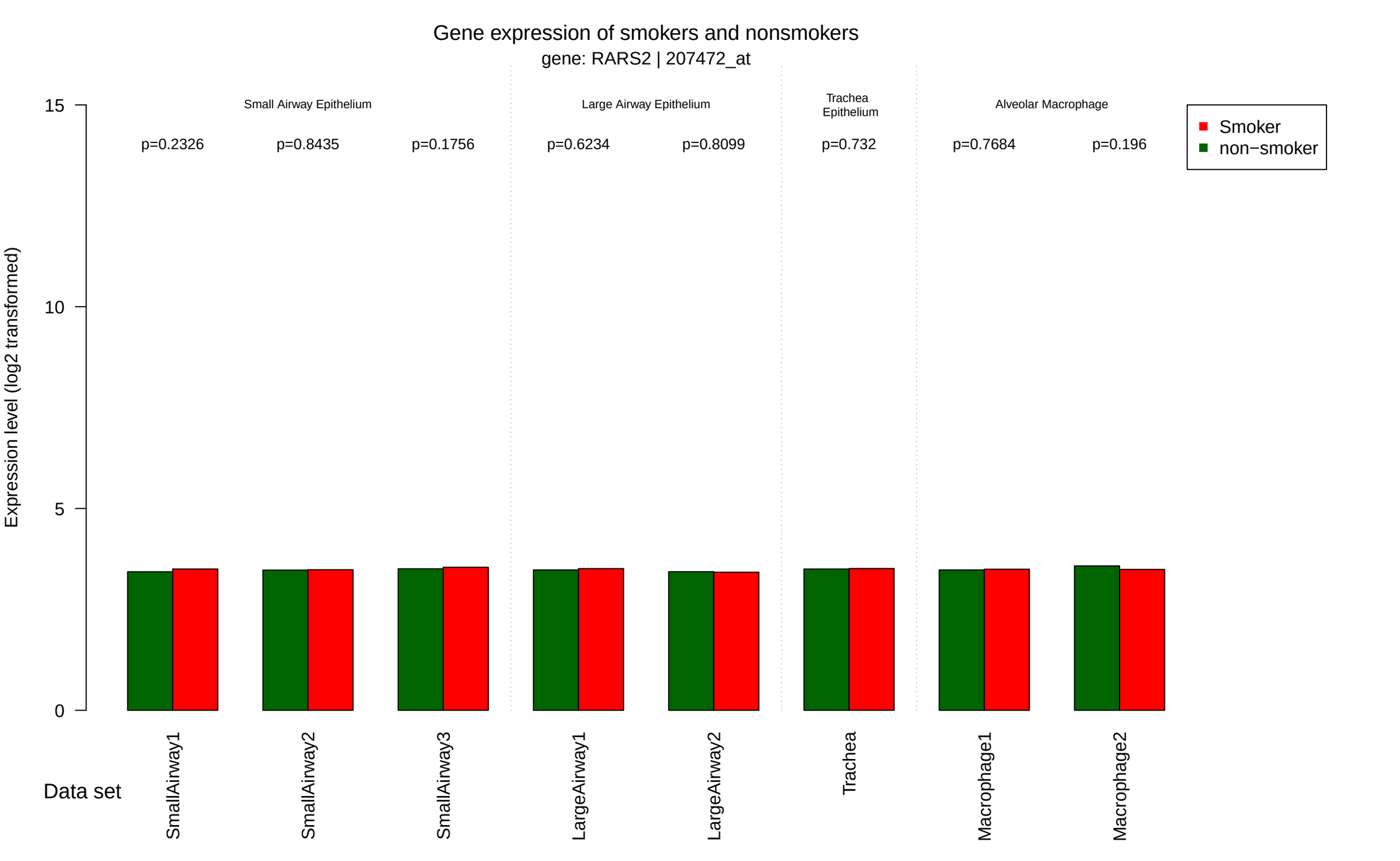
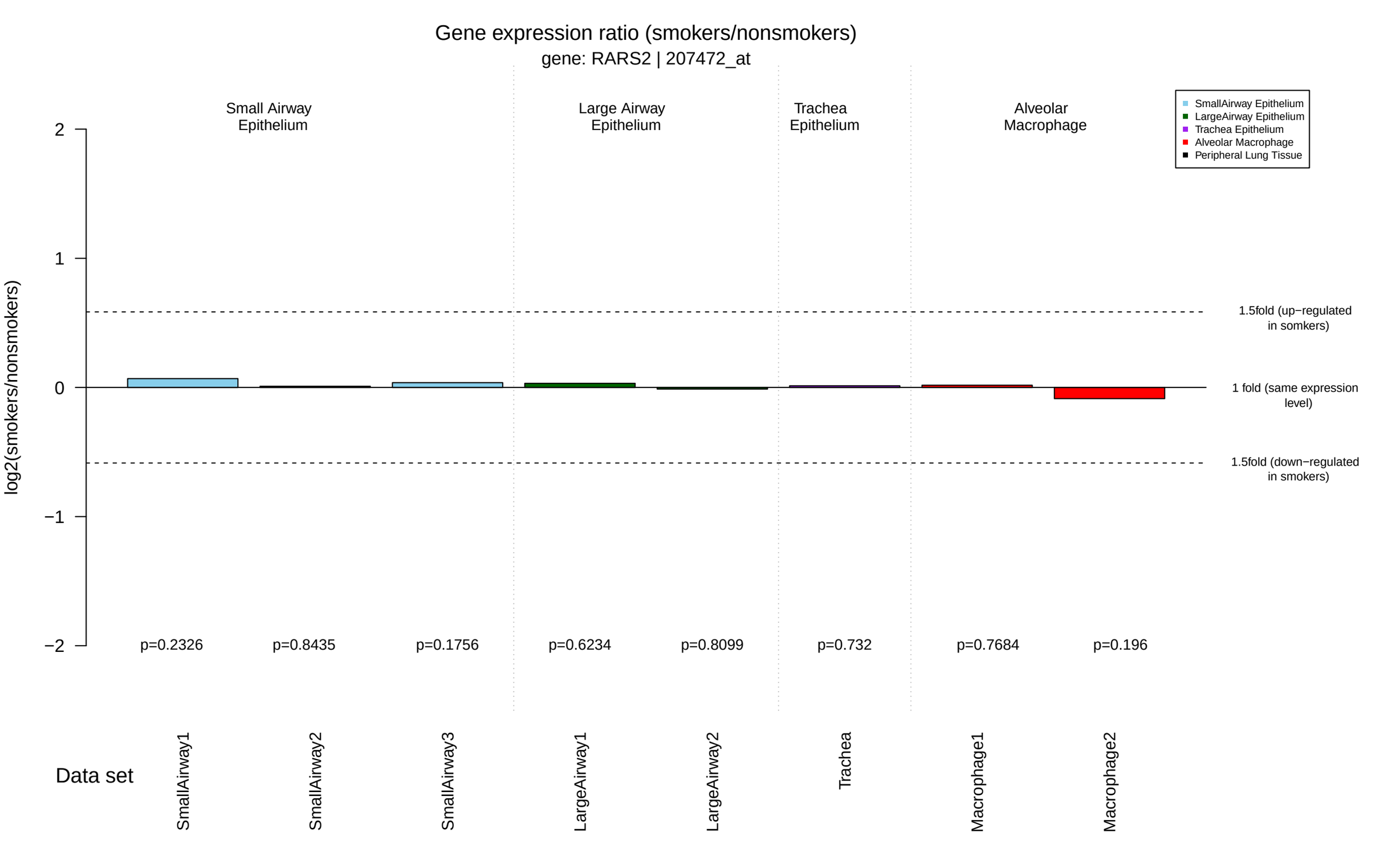
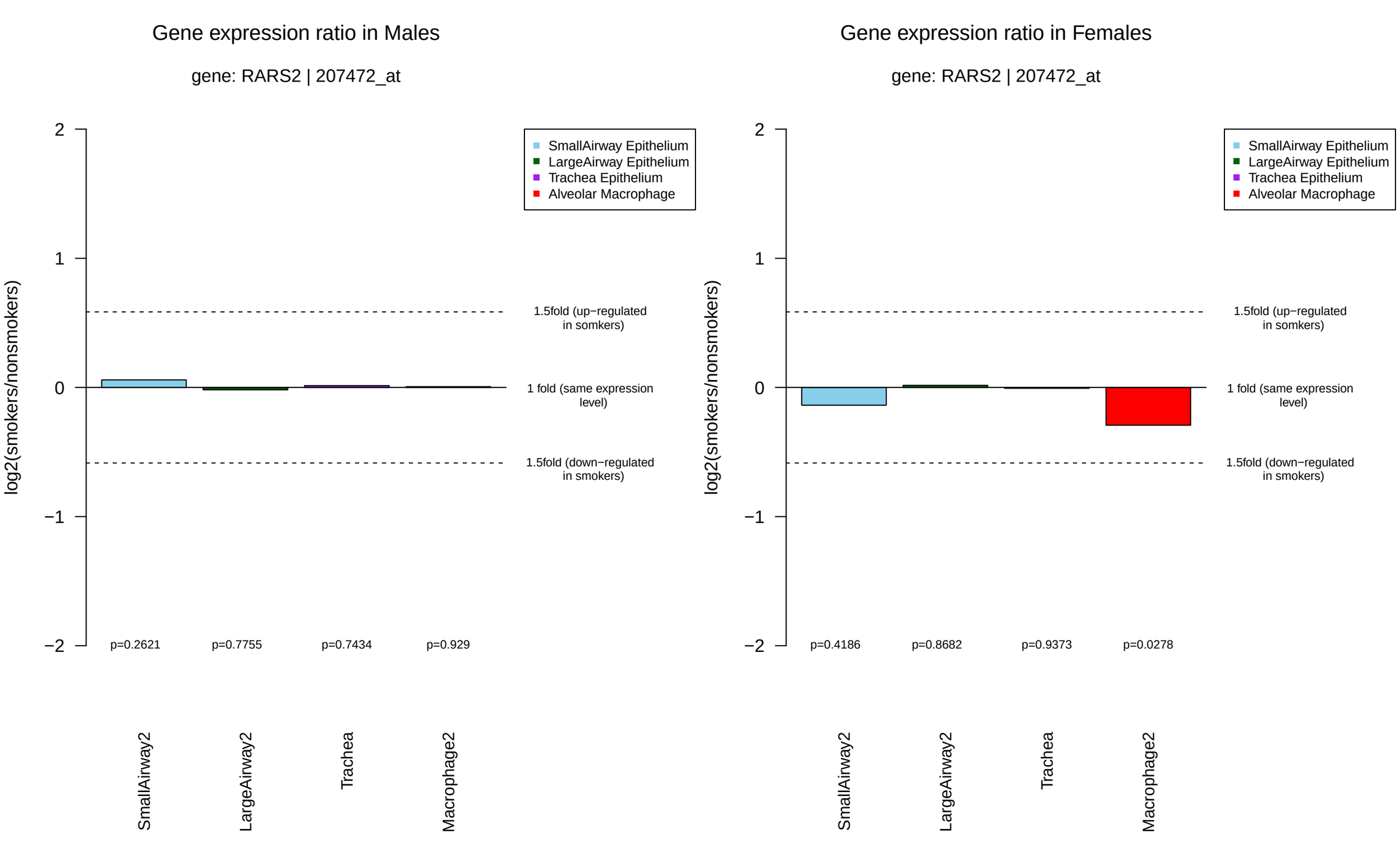
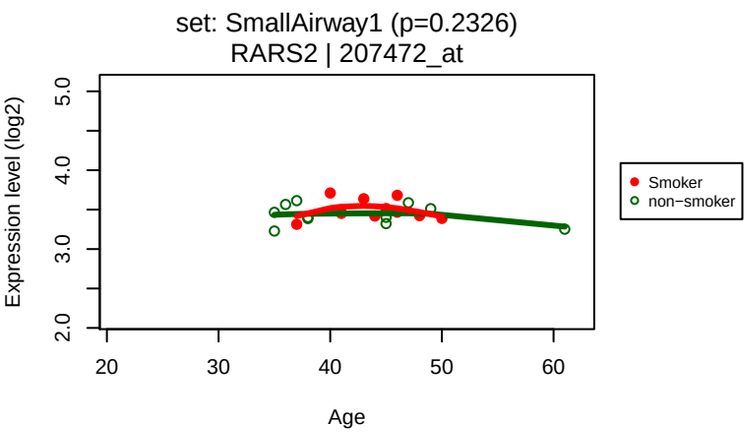
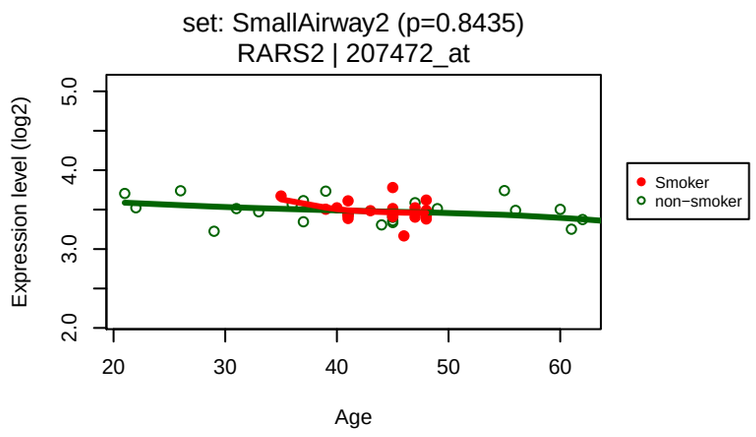
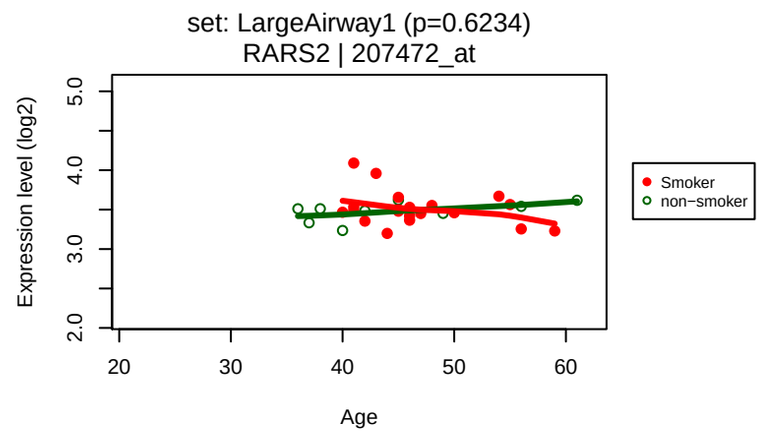

Gene expression of smokers and nonsmokers
gene: RARS2 | 207472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2326
p=0.8435
p=0.1756
p=0.6234
p=0.8099
p=0.732
p=0.7684
p=0.196
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

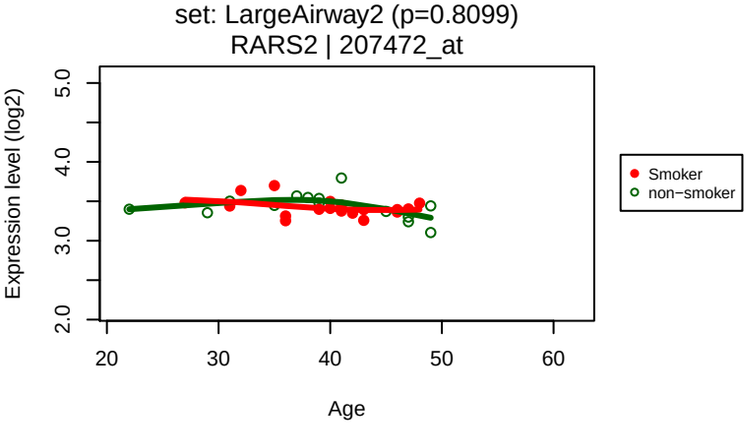

Gene expression ratio (smokers/nonsmokers)
gene: RARS2 | 207472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2326
p=0.8435
p=0.1756
p=0.6234
p=0.8099
p=0.732
p=0.7684
p=0.196
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

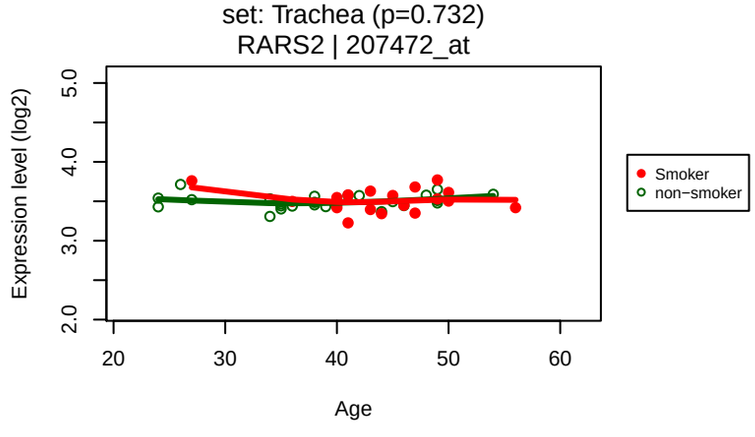

Gene expression ratio in Males
gene: RARS2 | 207472_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2621
p=0.7755
p=0.7434
p=0.929
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RARS2 | 207472_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4186
p=0.8682
p=0.9373
p=0.0278
SmallAirway2
LargeAirway2
Trachea
Macrophage2

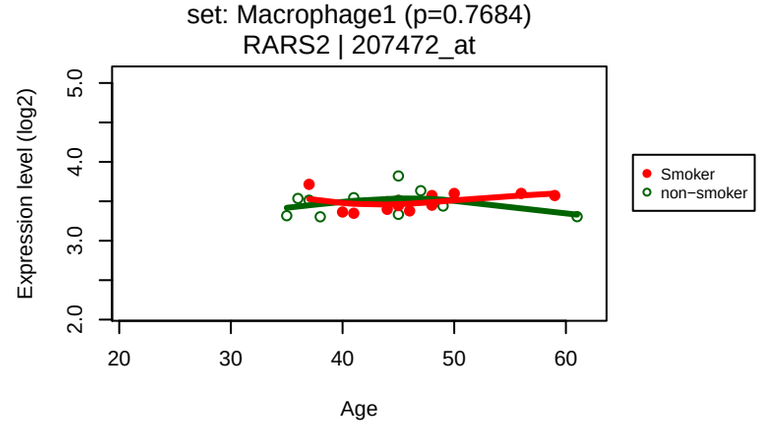

set: SmallAirway1 (p=0.2326)
RARS2 | 207472_at
Expression level (log2)
Age
Smoker
non−smoker

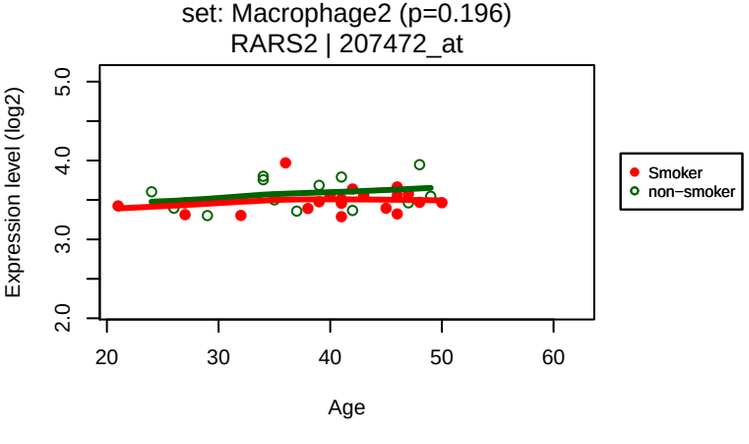

set: SmallAirway2 (p=0.8435)
RARS2 | 207472_at
Expression level (log2)
Age
Smoker
non−smoker

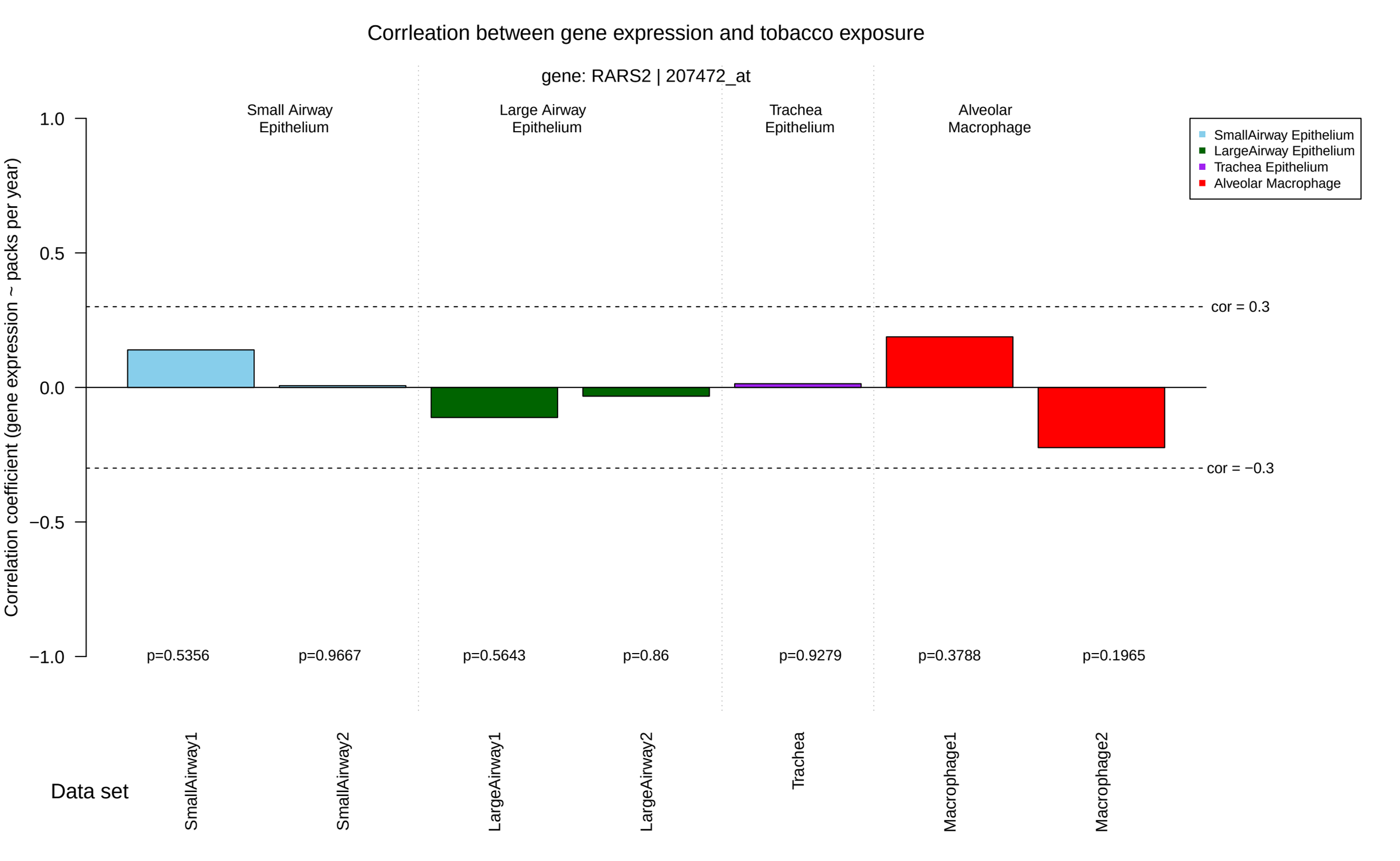

set: LargeAirway1 (p=0.6234)
RARS2 | 207472_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8099)
RARS2 | 207472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.732)
RARS2 | 207472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7684)
RARS2 | 207472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.196)
RARS2 | 207472_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RARS2 | 207472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5356
p=0.9667
p=0.5643
p=0.86
p=0.9279
p=0.3788
p=0.1965
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2